IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Digital Object Identifier

Supplementary Material for Physics-Informed Bayesian Event Attribution From Sparse PMUs

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ABSTRACT This supplement gives the executable statistical contract of the causal physical-history method: local DAE reduction, candidate transitions, truncated-severity evidence, state updates, restoration, worked examples, and scenario-level reproduction. It also defines the measurement-integrity interface and separates causal alarm, diagnosis and restoration metrics from the historical hackathon window evaluation. Earlier reconstruction, batch-source and proposed joint-estimator studies are preserved in the separately compiled Historical Technical Notes. Their results do not enter the causal method's denominators.

INDEX TERMS phasor measurement units, physical event attribution, measurement integrity, causal inference, reproducibility.

APPENDIX A HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections A–C give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal physical history bank. The conditional Gaussian updates use standard filtering identities [1]; the derivation below specifies their role and the approximations made by the implemented bank.

A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the frozen method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

B. EXECUTION ORDER FOR ONE STREAM

1) **Load the frozen operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.
3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive den-

sity before correcting states or choosing new histories with the current report.

4) **Propose candidate onsets.** The frozen screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode closure retain the augmented-state mixture mean; the omitted between-history spread is a documented approximation, not an exact posterior covariance.
7) **Publish and record.** Aggregate active mass M, cause mass Γ and ΔM. Apply the frozen detection, diagnosis and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{1}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

D. FROZEN NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

The internal observer confirms active-mass evidence at 0.99 for six reports and has a separate 15-report internal OFF clock. The public physical-bank publisher requires an internal pending episode and six reports of source-qualified active support above 0.99; its public OFF requires six reports of the diagnosed cell's joint cause-and-restoration mass above 0.99. The detailed source qualification, diagnosis and replacement rules are given below. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their frozen publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

E. SOURCE QUALIFICATION AND PUBLIC EPISODE RULES

The source screen and the onset search are distinct. Source qualification uses a 30-report (one-second) conditional innovation vector v with 47 coordinates per report. Fixed normal-mode gains propagate the stored pre-window state and whiten the suffix. For each prepared response s_j and lag from 0 to 25 reports, let

$$\begin{aligned} g_j &= s_j^\top s_j, \qquad u_j = s_j^\top v, \qquad R = v^\top v, \\ \widehat{a}_j &= \mathrm{clip}(u_j / \max(g_j, 10^{-30}), -0.2, 0.2), \\ G_j &= 2\widehat{a}_j u_j - \widehat{a}_j^2 g_j, \\ F_j &= \frac{\max(G_j, 0)}{\max\{(R - G_j)/1409, 1\}}. \end{aligned} \tag{2}$$

Line columns use fixed amplitude one. The source score is the maximum over its columns; the threshold is the $1 - 0.001/936$ quantile of an $F_{1,1409}$ distribution, approximately 24.0115. This threshold and the variance floor are a frozen proposal/publication rule, not a calibrated global false-alarm probability. The statistic is never multiplied into the Bayesian history likelihood. This one-second qualification screen does not enlarge the separate birth search beyond its 0–18 report lags.

Define $b_c = 1$ for a line, and $b_c = \mathbf{1}\{F_c > 24.0115\}$ for a load. At each monitored report, the publisher updates consecutive-report counters for $M_c b_c \geq 0.99$, $\Gamma_c \geq 0.99$, and $\Gamma_c P(\mathrm{OFF} \mid c) \geq 0.99$. A false condition clears its counter. An internal alarm opens a pending episode. A public alarm then requires either six pending reports with $\sum_c M_c b_c \geq 0.99$ or an already accumulated six-report line-active counter. The latter permits line evidence collected before the internal pending message to count.

A public episode receives one diagnosis when its highest cause-mass cell has six consecutive reports above 0.99, or at the 15-report deadline after the public alarm. The output family maximizes summed cause mass; sources are ranked within that family. The diagnosed source becomes the episode owner. Six reports of that owner's joint cause-and-OFF mass above 0.99 emit a public restoration. These publication operations do not rescale M or reset the physical state.

The internal mixture lifecycle separately replaces an episode after six reports with mass at least 0.99 on active

histories whose onset is later than its alarm (using the implementation's quarter-report timing guard). It withdraws an unconfirmed episode after the diagnosis delay when null mass has exceeded 0.99 for 15 reports. Its internal restoration requires cause mass at least 0.99 and conditional retired mass at least 0.99 for 15 reports. These terminal messages are propagated to the public episode if already emitted; an unconfirmed pending episode is recorded as rejected. Finalization marks any remaining public episode right-censored. The nominal comparator retains its own fixed internal lifecycle, so the nominal/mixture comparison is not an isolated model-averaging ablation.

APPENDIX B LOCAL CALCULATIONS AND WORKED EXAMPLES

A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here μ, v are parameters of the underlying untruncated Gaussian, not the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$\begin{aligned}(e - sa)^\top S^{-1}(e - sa) + (a - \mu)^2/v \\ = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a - \mu')^2/v', \quad (3)\\ v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \quad (4)\end{aligned}$$

Define $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l, u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1} e + \log|S| + n_y \log(2\pi)] \\ & + \tfrac{1}{2}[\log(v'/v) + \mu'^2/v' - \mu^2/v] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \quad (5)\end{aligned}$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu', v')$. With $\alpha = (l - \mu')/\sqrt{v'}$, $\beta = (u - \mu')/\sqrt{v'}$, $Z' = \Phi(\beta) - \Phi(\alpha)$ and standard normal density ϕ, its mean is

$$\mathbb{E}[a \mid e] = \mu' + \sqrt{v'}\,\frac{\phi(\alpha) - \phi(\beta)}{Z'}. \quad (6)$$

Thus neither the untruncated mean μ' nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S = 1$, $s = 2$, $e = 1$, $\mu = 0$, $v = 1$ and $[l, u] = [0, 1]$. Then $g = 4$, $q = 2$, $v' = 0.2$ and $\mu' = 0.4$. Substitution in (5) gives $L \simeq 0.342688$ and $\mathbb{E}[a \mid e] \simeq 0.464941$. The different posterior mean follows from truncation. The script scripts/verify_supplement_examples.py checks the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a frozen experiment.

C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h + d_h a$. Let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With shared covariance P_ρ in discrepancy regime ρ,

$$\begin{aligned}S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \quad (7)\end{aligned}$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h + d_h a) + g_o + u_o a + K_o(e - sa) = m_h^+ + d_h^+ a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. The physical history bank does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x} = -x + u$, $y = x$, with $x(0) = 0$. Apply $u = a$ for $0 \le t < \kappa$ and restore $u = 0$ afterwards. Direct integration gives

$$x(t) = \begin{cases} a(1 - e^{-t}), & 0 \le t < \kappa, \\ a(1 - e^{-\kappa})e^{-(t-\kappa)}, & t \ge \kappa. \end{cases} \quad (8)$$

Both expressions agree at κ. For $a = 1$ and $\kappa = 2$, $x(2) = 0.864665$ and $x(3) = 0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c = 0.60$, while its originating-cause mass is $\Gamma_c = 0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k} M_{\ell,c,k-1}$ proves

$$\begin{aligned}\Delta M_c = & \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ & + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \quad (9)\end{aligned}$$

For example, unchanged conditional masses $(0.8, 0.2)$ and network weights changing from $(0.5, 0.5)$ to $(0.75, 0.25)$ move M_c from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

APPENDIX C SIMULATION SCENARIOS AND REPRODUCTION ORDER

A. PLANT, MEASUREMENT AND PROVENANCE

The active simulator is the frozen research candidate `SIM0001_JuliaPD_F01`, not an officially validated equivalent of RAW0001. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the frozen impedance law.

The voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. Each current is an oriented branch-terminal measurement, not a nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. RAW0001 provided retrospective calibration context. RAW0002 remains outside these studies.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t = k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not interpolated across a switch. The causal instrumental history is preserved through OFF.

B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

H03 and H04 each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

The confirmatory evaluation prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18+27+12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. Completed results appear in the dedicated confirmation tables.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. Chen2016 is eligible on that lane; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The E_i/X02 signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The X04 recovery geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate the causal physical method, unknown onset, event cardinality or operational false alarms.

D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's frozen plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the F01 state, parameter and measurement files from its snapshot. For any new simulation create a new run directory; a new hypothesis requires a successor candidate. Do not run historical preparation scripts inside a frozen baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures with scripts/build_manuscript_bundle.ps1. Its default compilation does not regenerate scientific artifacts or import new campaign results. An explicit artifact refresh is a separate operation.

The repository keeps the current walkthrough in this appendix, the frozen campaign evidence under `generated/frozen/`, and the earlier mathematical development in the separately compiled Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

E. F01 MEASUREMENT CONTRACT

The executed channel map is taken from the frozen SIM0001-JuliaPD-F01 research snapshot. The eight voltage buses are

$\{2, 5, 6, 10, 19, 22, 29, 39\}$. Voltage magnitude is reported in per unit with unit gain; voltage angle uses one common affine phase reference plus the archived relative offsets. The reference interpretation is observational: physical common drift and instrument clock drift are not separately identified. Terminal currents are converted to outward branch convention from the simulator's into-terminal observable, then mapped with the row-specific polarity, gain and phase bias. Current bases use $100\,\mathrm{MVA}/(\sqrt{3}\,V_{\mathrm{terminal}})$; the common voltage base is 199.186 kV.

Frequency is computed as $60 + \dot{\theta}/(360)$ from the reported voltage angle, and ROCOF as $\ddot{\theta}/360$ using a causal 21-report quadratic estimate. The generator fit is robust and magnitude weighted; the predictor uses a fixed unweighted quadratic fit over the same history. Four warmup outputs are unavailable; the confirmatory evaluation uses a 10-s startup context and begins monitoring after 12 s. The 47 retained coordinates exclude the BUS39 current-angle coordinate. Noise multipliers 1 and 2 scale the standard deviation under the frozen stream. The stochastic model archives a cross-signal covariance field, but the executed report operator applies noise by signal family without that cross-family factor. These choices describe a simulated sensor configuration, not a verified eight-PMU CT/PT installation.

F. EVALUATED DOMAIN PROFILE

The confirmatory evaluation covers 18 continuous six-event parent schedules, repeated under three network conditions and two standard-deviation noise scales, for 108 120-s event records per method (3.6 event-record hours). It includes 12 normal parents under the same condition/noise crossing, giving 72 600-s records (12.0 planned normal hours). A separate persistent-line lane has nine line sources with three repetitions each, crossed with the same network/noise settings. The twelve-second monitoring warmup follows a ten-second context fit; the earliest scheduled intervention is at 16 s.

Across the 18 unique continuous schedules, finite event duration ranges from 2.02 to 4.95 s. Load changes multiply both active and reactive setpoints by $1 + a$ under the frozen impedance law; the prespecified $|a|$ range is 0.03–0.14 (realized signed schedule values range -0.138 to 0.135). Line events are full branch openings followed by restoration. The three network conditions are MATCH, RX_IN_SUPPORT and RX_OUT_SUPPORT; out-of-support cases use independently signed 1–3

Noise scales 1 and 2 are standard-deviation multipliers of the frozen R5C.2 stream, with the same seed paired across conditions and scales. Event extras retain the full 120-s event-record denominator; this is not the monitored post-warmup duration. Parent schedules, not repeated condition/noise runs or PMU samples, are the independent unit. The catalog contains nine candidate loads and nine candidate lines; it is not an unseen-source or all-39-bus evaluation.

APPENDIX D MEASUREMENT INTEGRITY AND THE HISTORICAL SUBMISSION INTERFACE

An electrical intervention and a measurement failure are separate objects. The physical output identifies a family and asset; the integrity output identifies missing or corrupt PMU observations. Missingness alone does not establish malicious intent. A concurrent physical event and missing PMU can therefore have two different locations.

The successor interface reuses the fitted detector, classifier and localizer from the actual submitted archive, rather than a similarly named exploratory pipeline. Archive member hashes are checked before inference. The input schema admits timestamps, three-phase voltage/current magnitudes and angles, frequency, ROCOF and the observed availability flag. Event labels, locations and annotated ON/OFF times are excluded. The fitted feature lists are also checked for label fields. This distinction matters because a historical diagnostic utility estimated a baseline from labeled normal samples over the complete record; that utility is not part of this inference interface.

The submitted classifier uses a physical-family head, a bad-data head and a missing/composition head with fixed rules. Its E5 and E7 outputs refer to missing and bad-data PMUs, respectively. E6 denotes missing data together with a physical prediction; its legacy location is the physical source, not the missing PMU. In the archived configuration, both composition and physical-confidence thresholds are zero, so E6 requires a nonzero physical prediction in a window with missing data; it is not a calibrated test of concurrent causation. E8 is the legacy mixed-physical proxy class, not an open-set rejection guarantee.

The availability interface evaluates each received report without looking ahead. It retains the affected PMU and channel mask rather than replacing missing measurements with future samples. The historical bad-data classifier uses a complete window and reports its result at the end. Within-window interpolation and centered features are permitted only under that delayed decision contract. Broadcasting the result to earlier timestamps does not make those earlier labels causal. The inherited cyber/integrity component is a comparator and integration component, not a newly claimed detector.

The implemented composition has two temporal scopes. Received-report availability messages update the current missing-PMU state. Completed-window records describe only their observed intervals; they retain separate physical and integrity fields and cannot assert current corruption or physical restoration. Their joint emission is the later component emission, with each original time retained. For example, a window that contains missing data can finish after the PMU has resumed reporting; publishing that interval must not reopen the missing episode. Likewise, an earlier ML source diagnosis does not advance the submitted classifier's window-completion time.

The sealed stream contains 206 messages, including 180 completed windows. Post-seal projection checks reproduce the physical source, integrity diagnosis, source-selection

TABLE 1. Eight PMU rows in the frozen F01 measurement map. V is a bus voltage in p.u.; I is an oriented branch-terminal phasor. Gains and phase biases are applied to current channels. BUS5 retains the historical 5–6, terminal-5 hypothesis and its producer CT identity is unverified. BUS39 current phase is partially validated.

PMU	V bus	Edge	Branch	Terminal	Side	Polarity	Gain	Bias (°)	I unit	Provenance
2	2	2	2–3	3	dst	1	0.9581	-1.173	A	F01 archived terminal hypothesis; CT unverified
5	5	9	5–6	5	src	1	1.0361	0.312	A	BUS5 5–6 historical hypothesis; CT unverified
6	6	11	6–7	6	src	1	1.0077	1.495	A	F01 archived terminal hypothesis; CT unverified
10	10	17	10–11	11	dst	-1	1.0041	2.224	A	F01 archived terminal hypothesis; CT unverified
19	19	26	16–19	19	dst	1	1.0384	0.481	A	F01 archived terminal hypothesis; CT unverified
22	22	34	21–22	22	dst	1	1.0189	0.094	A	F01 archived terminal hypothesis; CT unverified
29	29	44	28–29	29	dst	-1	1.0316	-6.170	A	F01 archived terminal hypothesis; CT unverified
39	39	1	1–39	39	dst	-1	0.9298	30.053	A	BUS39 partial phase validation

provenance and unavailable states of the components, and recover current availability on all 161379 reports without disagreement. The comparison retains the original component evaluations. It neither reruns inference nor creates a combined accuracy from report-level missingness and window-level physical diagnosis. The frozen physical filter remains explicitly unavailable.

APPENDIX E ALARM, DIAGNOSIS AND RESTORATION ARE DIFFERENT DECISIONS

Let τ and ω be true ON and OFF, and let t_A, t_D and t_R denote the emission times of an alarm, source diagnosis and restoration. The detector can signal before it has a source. An eventual correct source at $t_D > t_A$ is not a correct location already available at t_A. Timely attribution requires a correct family and source at $t_D < \omega$. Estimated onset times never replace emission times in a delay metric.

An event is associated with the first eligible unmatched alarm using time, before inspecting the predicted label. Incorrect labels remain associated errors rather than being reassigned to another event to improve the score. An alarm without a diagnosis, a wrong diagnosis, and no alarm are distinct outcomes. Restricted delay includes failures up to the evaluation horizon; conditional delay among successes is reported separately.

A restoration of an unmatched episode and a premature restoration of a matched event are also different errors. With a tolerance δ fixed before scoring, premature restoration is $t_R < \omega - \delta$; its lost active duration is $[\omega - t_R]_+$. Signed restoration delay is $t_R - \omega$. Report its distribution, missing closures and episodes still open at the horizon; a median alone can hide early closures.

The original evaluator is preserved. Its false-restoration counter counts restorations of unmatched episodes, and its first-correct field can use the diagnosis associated with the first alarm even when emitted later. The successor evaluation reports these quantities with their restricted meanings and adds explicit premature closure and joint-decision metrics. This is a correction of evaluation semantics, not a change to the frozen detector or evidence that either failure occurred in a particular study record.

APPENDIX F RETROSPECTIVE RAW COMPARISON AND INDEPENDENT UNITS

The archived RAW0001 table contains 21 contexts, including nine labeled-normal and twelve abnormal contexts. Its recovered loader places a 30-s window three seconds before a supplied abnormal onset, and selects normal contexts from supplied normal intervals. Even a successful reproduction would measure classification and localization within event-guided contexts, not continuous detection. However, the recovered calendar is incompatible with the event annotations in the actual RAW files. File-history checks show that those RAW files have not changed, and the timing differences cannot be corrected by a common offset. The archived score is therefore not a reproduced comparison endpoint. The unsuccessful reconstruction and its provenance audit are retained; predictions from mismatched windows are not counted as accuracy against the archived labels.

A separate complete-record replay uses fixed windows without event-guided placement and retains all normal, physical, missing and corrupt periods. Both interfaces preserve availability failures and abstentions. Physical and integrity locations are scored separately where their meanings differ; the legacy single-label export is retained for comparison. Physical-source scoring additionally requires verified source annotations; an event-code sequence alone cannot identify every intervention boundary or source. The separately sealed successor analysis uses the recovered original source workbook for physical identity, retaining current event-code support for window membership. Its coarse timing does not resolve the individual actuation or restoration times. This replay is not an official competition score. RAW0001 previously informed F01 calibration, so even a frozen new replay remains retrospective.

Simulation partitions use the complete physical parent and all of its derived windows, conditions and noise realizations. Validation belongs inside the development fraction. Repartitioning previously inspected campaigns does not restore independent-test status. Uncertainty calculations resample physical parents, retaining their paired conditions together. Different PMU channels and adjacent 30-Hz reports are not independent trials.

APPENDIX G REPOSITORY BOUNDARIES AND REPRODUCTION

The article and current supplement compile independently from `main.tex` and `supplement.tex`. The separate `historical_notes.tex` preserves corrected geometry and evidence theorems, earlier state reconstruction, the historical batch estimator, the proposed joint estimator and their original empirical qualifications. Moving this material does not remove negative results or upgrade an unexecuted estimator to an implemented one.

The submission successor is versioned in the research repository under experiments/PAPER_SUBMISSION_S01/. It contains the source provenance, observation-only interface, metric specification, synthetic regressions, and separate historical-window and complete-record outputs. Performance tables enter the manuscript only after prediction sealing and evaluation. A successful code check or PDF build is not itself a performance result.

APPENDIX H EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - as_k$. The affine conditional state is $m_k + ad_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{10}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{11}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{12}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a}U - \tfrac{1}{2}\widehat{a}^2 G - C_{ref}. \tag{13}$$

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

TABLE 2. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c)P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The H04 comparison contains no such post-startup missing observations.

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

APPENDIX I GROUPED LEARNING AND INTEGRITY PUBLICATION

A. RETROSPECTIVE SIM TRAINING CONTRACT

The SIM11–19 registry contains 176 observation records from 88 physical cases. Connected components join shared physical histories and observation replicas before splitting. Of 56 dependency groups, 31 are used for fitting, 8 for model selection and 17 for the reserved development evaluation: 90, 16 and 70 records, respectively. This implements the requested approximately 70/30 development/reserved partition with an internal validation subset. All campaigns were already open; the split is retrospective and does not establish unseen-source or external generalization.

Both ExtraTrees variants use 400 trees, minimum leaf size two, square-root feature sampling, balanced class weights and seed 2601026. Fitting uses every third report starting

at index 360. The target is normal outside an intervention and one of 18 load/line sources during it; recovery is not a separate training class. The baseline has 846 trailing features; SHORT adds 611 features. Their input is the same 47-channel representation with a 300-report empirical reference, colored filtering and median/MAD scaling. No DAE fit or RAW annotation enters fitting. The empirical reference has no startup stationarity guarantee; it is distinct from the physical filter's admission gate.

The publication adapter requires six consecutive winning source labels. Scores are not claimed to be calibrated probabilities. Selection maximizes validation diagnoses with correct family and source emitted before OFF, then minimizes extra alarms, preferring the baseline on a complete tie. SHORT is frozen after 108/192 versus 92/192 validation successes. There is no refit on validation and no subsequent parameter or threshold selection from the reserved 70 records. The declared metric associates the first temporal alarm independently of its predicted class, within 30 seconds or before the next onset, whichever comes first; success also requires diagnosis before the current OFF.

The reserved result is 302/576, comprising 28/288 load and 274/288 line interventions, with 3 extra alarms. Only two distinct normal dependency groups exist: one in training and one reserved; validation has none. The 3696 normal seconds are repeated observations of the reserved normal group, with 0 alarms. They do not provide independent long-duration operational exposure. In particular, neither report-level binomial intervals nor a validation false-alarm rate would be justified. Load weakness and the common F01 ancestry limit transfer claims even when the group assignment is disjoint.

The selected model is also replayed on all twelve H04 records, using the same successor event metrics as the physical controls. It obtains 27/54 timely decisions, 3 extras, 3 orphan restorations and 27 missing closures, including missed events. All missed events are loads; all tested lines are diagnosed before OFF. Its six-argmax publisher has no confidence threshold, so this is not an identical decision-clock ablation. The three H04 event groups are disjoint from the SIM fitting, validation and reserved groups, but its single normal group is shared with the reserved group. The inference-only replay reads no truth, registry or split. An earlier roster contained unused event/normal metadata; it is preserved and documented. Removing that metadata reproduces every output array and event exactly on all twelve records, before separate post-seal scoring.

B. MISSING-PMU NOTICE TIMING

The continuous RAW replay contains 4 observed missing intervals. The exact submitted artifact and the immediate availability interface both identify all intervals and their affected PMU. First correct notices have mean delay 12.13 versus 0 seconds, respectively. The clock is the received report timestamp relative to the first affected report; zero means emission on that report, not zero computation or communication latency. Notices before onset cannot match a later missing interval, and a later correct source cannot repair an incorrect first notice. This is a publication benefit from already-observed flags and finite-value masks, not evidence of a new malicious-attack detector or improved physical-source inference.

C. SEGMENTED ML TRANSFER AND SOURCE COMPOSITION

The classifier consumes the phase-A voltage/current magnitudes and angles, frequency and ROCOF in the declared eight-PMU order. Complete segments require all 48 measurements finite and all presence flags true; missing reports are never removed to join their neighboring samples. Each segment gets its own 300-report empirical reference and remains unavailable through index 359. Segment startup is not certified normal and physical state is not preserved across gaps. The same frozen classifier and publication rule are used; no RAW source label, event code or score enters inference.

For a submitted-artifact prediction of line or load, including its predicted physical component in a mixed missing case, source replacement requires the first ML diagnosis of an episode with that same predicted family, emitted inside the completed window no later than the artifact's own emission. Later revisions cannot repair that first diagnosis. The earliest eligible diagnosis is selected; otherwise the ZIP source is retained. Fault, generation, integrity outputs and all window class labels and times are unchanged. The archived Top-3 list remains identified as the original ZIP list; it is not presented as the hybrid's reranked posterior.

The completed replay has 5 complete segments and 1 selected source substitution, with 179 fallbacks. The evaluator verifies full report coverage, explicit startup unavailability, causal message times and every replacement against the frozen rule before scoring. Source identity is supplied only to this later evaluator by the original metadata workbook: fault BUS39, line 23–24, generation BUS2 and load BUS7. Current event-code support determines which fixed windows contain each family; the workbook's approximate times do not supply exact action onsets or restorations. All physical-positive windows, including generation outside the learned catalog, enter the denominator. Paired joint correctness is 1/24 versus 2/24. These are dependent windows of one retrospective record; no population-level significance or official competition improvement is inferred.

APPENDIX J DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete F01 trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event

trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate F01 measurement operator is followed by its frozen innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 1). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an improvement in the causal physical bank nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

APPENDIX K OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of the causal physical method.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{14}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the frozen common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2 Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{15}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The frozen F01 plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three frozen instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its archived instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from

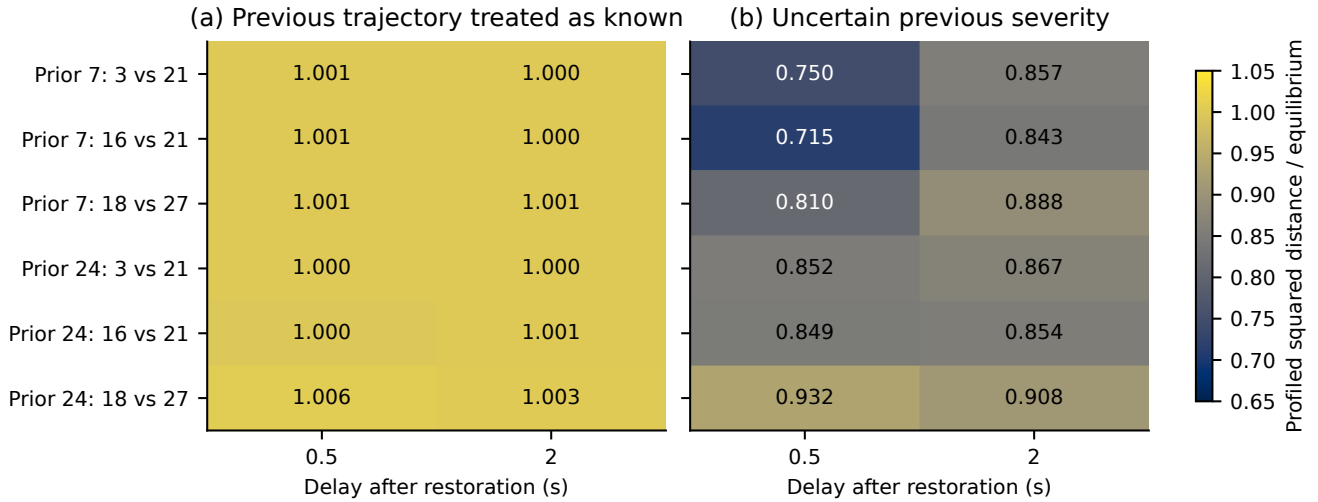


FIGURE 1. **Full-F01 physical control, not a performance comparison for the causal physical bank. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

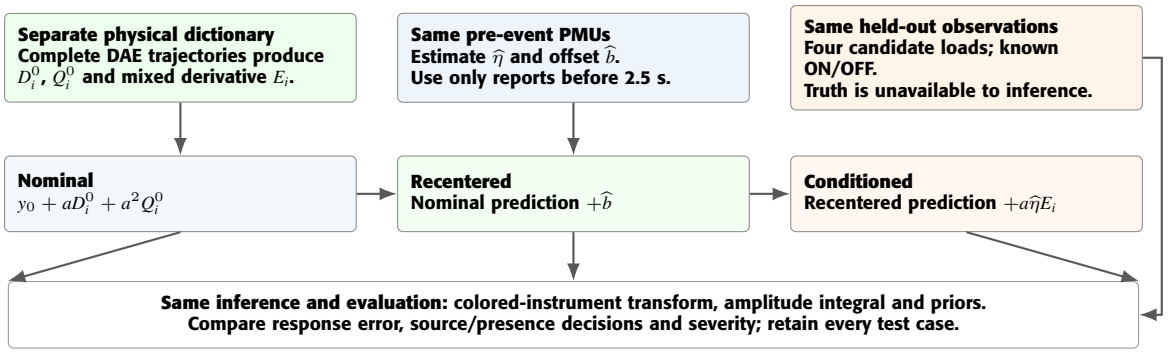


FIGURE 2. **Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.**

TABLE 3. **Controlled known-onset load test. Response error uses 80 physical cases; joint and source-only correctness use 240 noisy event records. Severity error is evaluated at the generating source. The 15 normal windows are not normal-hours exposure.**

Measure	Nominal	Recentered	Conditioned
Response error (%)	1.25	1.25	0.0561
Joint correct	114/240	240/240	240/240
Source-only correct	114/240	240/240	240/240
Normal selections	12/15	0/15	0/15
Severity error (%)	3.45	1.2	0.329

1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 3 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the frozen impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the frozen artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

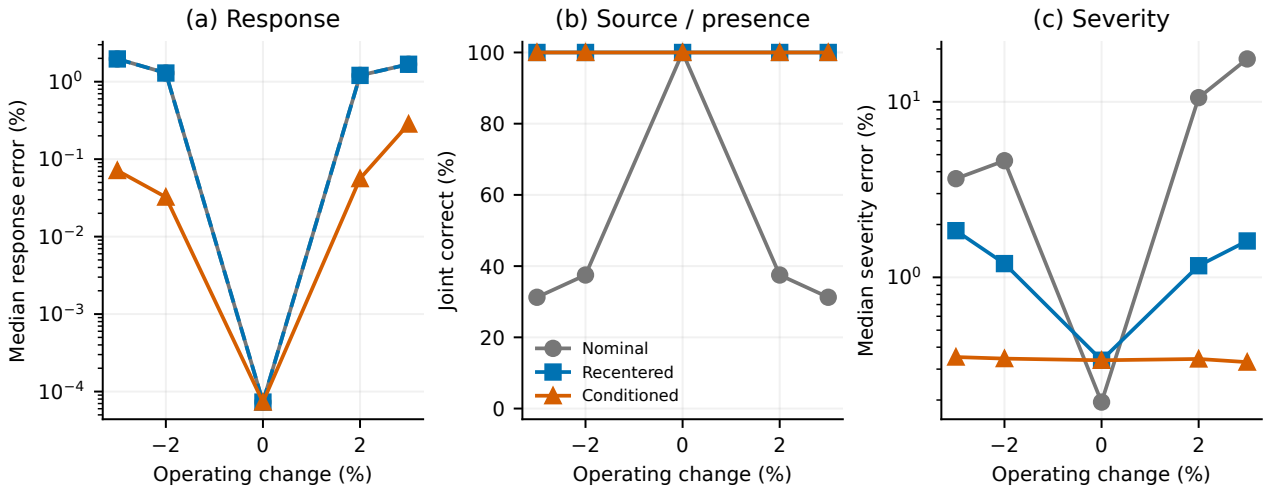


FIGURE 3. **Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.**

APPENDIX L COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The frozen test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated directly from the sealed summary, records, event matches and paired intervals.

A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 5 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 11 describes those already recorded outputs. It does not change the frozen first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective RAW0001 replay under its startup noise-scale contract.

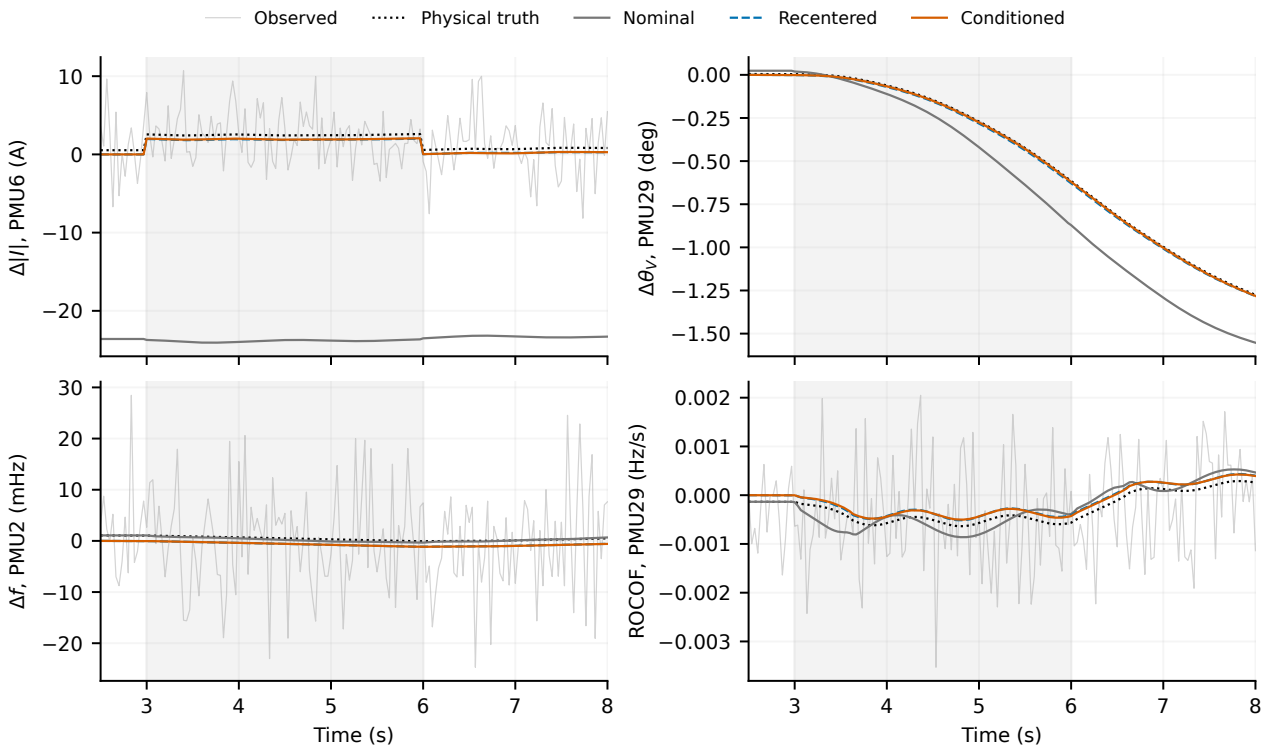


FIGURE 4. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

TABLE 4. **Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.**

Network	Noise	Physical bank	Nominal bank	ExtraTrees	ET short
Matched	1	108/108; 1	108/108; 1	56/108; 6	60/108; 5
Matched	2	108/108; 0	108/108; 0	54/108; 4	58/108; 0
In-bank	1	108/108; 2	108/108; 3	56/108; 5	60/108; 5
In-bank	2	108/108; 1	108/108; 2	54/108; 4	57/108; 0
Out-of-bank	1	106/108; 38	105/108; 47	56/108; 5	60/108; 6
Out-of-bank	2	106/108; 30	106/108; 39	54/108; 5	57/108; 0

TABLE 5. **Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, unmatched-episode restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.**

Method	Unavailable	Normal incidence	Normal alarms	Exposure (h)	Unmatched episode restorations	Open at end
Physical 5-model	0	0/72	0	11.760	58	0
Physical nominal	0	0/72	0	11.760	68	0
ExtraTrees	0	0/72	0	11.760	29	0
ET short	0	0/72	0	11.760	16	0

TABLE 6. Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.

R/X condition	Noise	Physical 5-model	Physical nominal	ExtraTrees	ET short
Matched	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
Matched	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
In-bank	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
In-bank	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
Out-of-bank	1	0.33/0.18	0.33/0.18	0.23/0.18	0.19/0.18
Out-of-bank	2	0.26/0.18	0.33/0.18	0.30/0.18	0.20/0.18

TABLE 7. Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the frozen evaluator.

R/X condition	Noise	Physical 5-model	Physical nominal	Chen2016
Matched	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Matched	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
In-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.04
In-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.05
Out-of-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Out-of-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.59

TABLE 8. Persistent-line coverage and stopping outcomes across six repeated strata, including both frozen ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen2016 normal exposure ends at its first stop; the other methods monitor the full available record.

Method	Unavailable	Correct first	Pre-change	Censored	Normal incidence	Exposure (h)
Physical 5-model	0	162/162	0	0	0/72	11.760
Physical nominal	0	162/162	0	0	0/72	11.760
ExtraTrees	0	162/162	0	0	0/72	11.760
ET short	0	162/162	0	0	0/72	11.760
Chen2016	0	74/162	66	0	72/72	0.701

TABLE 9. Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.

Family / asset	Physical bank	Nominal bank	ExtraTrees	ET short
Line / LINE14-15	36/36	36/36	27/36	34/36
Line / LINE16-17	36/36	36/36	36/36	36/36
Line / LINE17-18	48/48	48/48	48/48	48/48
Line / LINE23-24	30/30	30/30	30/30	30/30
Line / LINE26-27	24/24	24/24	24/24	24/24
Line / LINE26-29	24/24	23/24	24/24	24/24
Line / LINE3-4	42/42	40/42	42/42	42/42
Line / LINE4-14	46/48	48/48	48/48	48/48
Line / LINE9-39	36/36	36/36	36/36	36/36
Load / BUS15	42/42	42/42	6/42	6/42
Load / BUS16	42/42	42/42	0/42	3/42
Load / BUS20	12/12	12/12	6/12	12/12
Load / BUS24	42/42	42/42	0/42	0/42
Load / BUS26	18/18	18/18	0/18	0/18
Load / BUS3	42/42	42/42	3/42	3/42
Load / BUS31	40/42	40/42	0/42	0/42
Load / BUS4	36/36	36/36	0/36	6/36
Load / BUS7	48/48	48/48	0/48	0/48

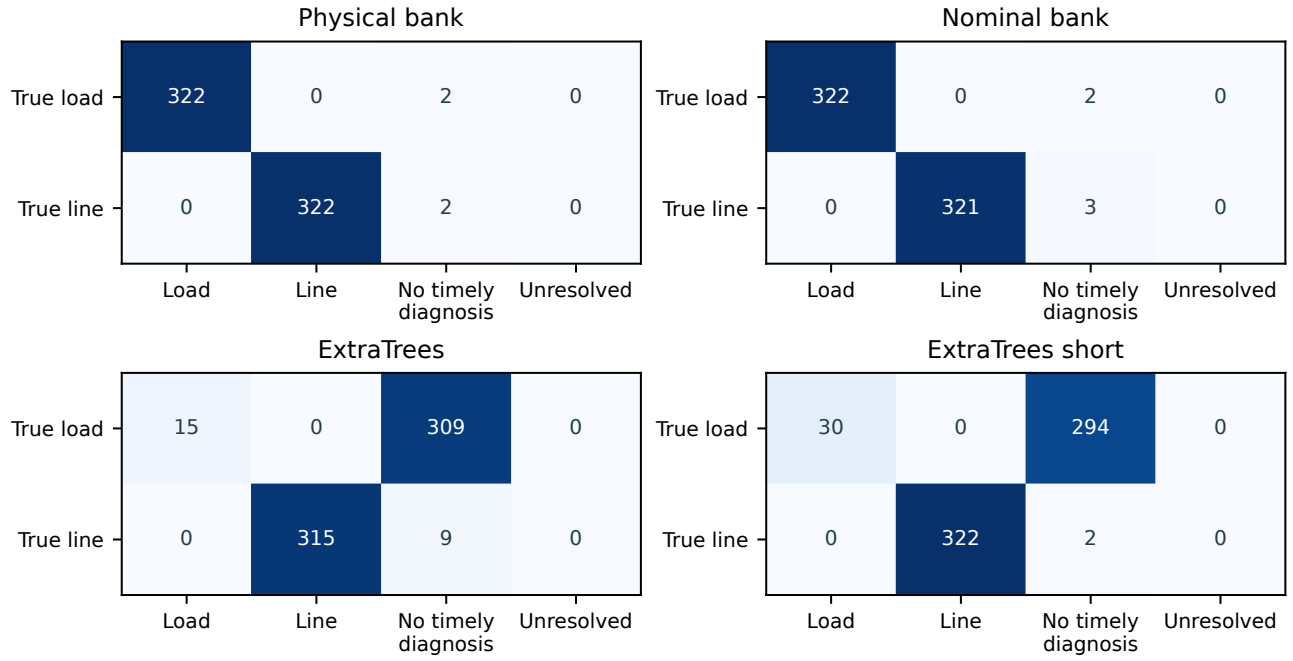


FIGURE 5. **Posthoc confusion audit of the frozen continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the prediction seal. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

TABLE 10. **Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per full event-record hour, including startup. Brackets are the frozen 95-percent parent-bootstrap intervals.**

Network	Noise	Method	Reference	Timely difference	Extra-rate difference
Matched	1	Physical bank	ExtraTrees short	44.44 [40.48, 47.92]	-6.67 [-15.00, 0.00]
Matched	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-8.33 [-18.33, 0.00]
Matched	1	Nominal bank	ExtraTrees short	44.44 [40.74, 48.04]	-6.67 [-13.33, 0.00]
Matched	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-8.33 [-18.33, 0.00]
Matched	2	Physical bank	ExtraTrees short	46.30 [42.75, 49.12]	0.00 [0.00, 0.00]
Matched	2	Physical bank	ExtraTrees	50.00 [46.30, 53.70]	-6.67 [-13.33, -1.67]
Matched	2	Nominal bank	ExtraTrees short	46.30 [42.86, 49.12]	0.00 [0.00, 0.00]
Matched	2	Nominal bank	ExtraTrees	50.00 [46.38, 53.70]	-6.67 [-13.33, -1.67]
In-bank	1	Physical bank	ExtraTrees short	44.44 [40.62, 47.92]	-5.00 [-13.33, 5.00]
In-bank	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-5.00 [-13.33, 5.00]
In-bank	1	Nominal bank	ExtraTrees short	44.44 [40.62, 47.92]	-3.33 [-11.67, 6.67]
In-bank	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-3.33 [-11.67, 5.00]
In-bank	2	Physical bank	ExtraTrees short	47.22 [43.18, 50.98]	1.67 [0.00, 5.00]
In-bank	2	Physical bank	ExtraTrees	50.00 [46.49, 53.70]	-5.00 [-10.00, 0.00]
In-bank	2	Nominal bank	ExtraTrees short	47.22 [43.33, 51.04]	3.33 [0.00, 8.33]
In-bank	2	Nominal bank	ExtraTrees	50.00 [46.30, 53.70]	-3.33 [-10.00, 3.33]
Out-of-bank	1	Physical bank	ExtraTrees short	42.59 [38.60, 46.43]	53.33 [36.67, 71.67]
Out-of-bank	1	Physical bank	ExtraTrees	46.30 [42.11, 50.00]	55.00 [36.67, 73.33]
Out-of-bank	1	Nominal bank	ExtraTrees short	41.67 [36.11, 46.67]	68.33 [46.67, 90.00]
Out-of-bank	1	Nominal bank	ExtraTrees	45.37 [41.11, 50.00]	70.00 [45.00, 95.00]
Out-of-bank	2	Physical bank	ExtraTrees short	45.37 [41.11, 50.00]	50.00 [35.00, 65.00]
Out-of-bank	2	Physical bank	ExtraTrees	48.15 [43.59, 52.50]	41.67 [25.00, 60.00]
Out-of-bank	2	Nominal bank	ExtraTrees short	45.37 [41.11, 50.00]	65.00 [41.67, 88.33]
Out-of-bank	2	Nominal bank	ExtraTrees	48.15 [43.75, 52.56]	56.67 [31.67, 80.00]

TABLE 11. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.

Method	Openings	Extras	Unmatched OFF	Open at end	FN reports
Physical bank	162	0	0	162	810
Nominal bank	162	0	0	162	810
ExtraTrees	162	164	164	16	335374
ExtraTrees short	162	179	178	61	182600

TABLE 12. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.

Component	Positive-window hits	FP	FN
Missing	14/14	0	0
Corrupt	1/2	2	1

APPENDIX M CONTROLLED COMPONENT ABLATION

The frozen protocol and all output tables are supplied in the source archive under generated/frozen/ablation_a01_20261102. Each of four arms accounts for all 24 intended records (18 continuous, three normal and three persistent): the evaluation audit records 96/96 available outputs and zero failures. The trial is posthoc and no score threshold is calibrated to these outcomes. Each normal control is a 600-s record with 588 s of evaluated exposure after startup, for 1,764 s total.

REFERENCES

[1] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.

TABLE 13. All outcomes, including restoration messages on unmatched episodes and episodes open at the record end.

method	lane	available	events	timely	extras	false restorations	open eof
Marginal + carry	continuous	18	108	108	1	1	0
Marginal + carry	normal	3	0	0	0	0	0
Marginal + carry	persistent_line	3	3	3	0	0	3
Marginal + reset	continuous	18	108	75	166	130	11
Marginal + reset	normal	3	0	0	0	0	0
Marginal + reset	persistent_line	3	3	3	0	0	3
Profile + carry	continuous	18	108	108	1	1	0
Profile + carry	normal	3	0	0	0	0	0
Profile + carry	persistent_line	3	3	3	0	0	3
Profile + reset	continuous	18	108	75	166	130	11
Profile + reset	normal	3	0	0	0	0	0
Profile + reset	persistent_line	3	3	3	0	0	3

TABLE 14. Paired descriptive differences with 95-percent physical-parent bootstrap intervals; no multiplicity-adjusted superiority claim.

method	reference	metric	difference	lower95	upper95
Marginal + carry	Profile + carry	timely_fraction	0.000	0.000	0.000
Marginal + carry	Profile + carry	extras_per_event_hour	0.000	0.000	0.000
Marginal + carry	Marginal + reset	timely_fraction	0.306	0.241	0.380
Marginal + carry	Marginal + reset	extras_per_event_hour	-275.000	-325.000	-223.333
Marginal + reset	Profile + reset	timely_fraction	0.000	0.000	0.000
Marginal + reset	Profile + reset	extras_per_event_hour	0.000	0.000	0.000
Profile + carry	Profile + reset	timely_fraction	0.306	0.241	0.380
Profile + carry	Profile + reset	extras_per_event_hour	-275.000	-325.000	-223.333